Unit 3 Homework — Equations & Inequalities — Answer Key

Complete independently · One/Two-Step · Distributive · Multi-Step · Special Cases · Inequalities · Word

A. One- & Two-Step Equations

Problem 1 Score 3

Solve: $5x = 45$

Answer: 9

Problem 2 Score 3

Solve: $2x - 7 = 9$

Answer: 8

Problem 3 Score 3

Solve: $\frac{x}{4} + 5 = 8$

Answer: 12

Problem 4 Score 4

Solve: $-3x + 2 = -10$

Answer: 4

B. Distributive Property & Variables on Both Sides

Problem 1 Score 3

Solve: $2(3x + 1) = 26$

Answer: 4

Problem 2 Score 4

Solve: $8x - 1 = 5x + 17$

Answer: 6

C. Multi-Step Equations

Problem 1 Score 4

Solve: $2(x + 5) = 3x - 1$

Answer: 11

Problem 2 Score 4

Solve: $10 - x = 2x + 1$

Answer: 3

Problem 3 Score 5

Solve: $6x + 2 = 3(x + 8)$

Answer: $\frac{22}{3}$

Problem 4 Score 5

Solve: $\frac{3}{4}(x + 8) - 1 = x + 2$

Answer: 12

D. Special Cases — No Solution or Infinite Solutions

Problem 1 Score 4

Solve: $3x + 7 = 3x - 4$ — how many solutions?

Answer: no solution

Problem 2 Score 5

Solve: 2(x + 6) = 2x + 12 — how many solutions?

Answer: infinitely many solutions

Problem 3 Score 5

Why does 4(x − 1) + 2 = 4x − 2 have infinitely many solutions?

A. After you simplify, both sides are exactly the same: 4x − 2.

B. The x-terms are the same, so there is no solution.

C. x = 0 is the only solution.

D. Any equation with 4x on both sides has infinitely many solutions.

Answer: A

E. Linear Inequalities

Problem 1 Score 3

Solve: 4x ≥ 32

Answer: x ≥ 8

Problem 2 Score 3

Solve: 2x + 7 ≤ 19

Answer: x ≤ 6

Problem 3 Score 4

Solve: −5x < 20 (careful!)

Answer: x > −4

Problem 4 Score 5

Solve: −4x − 3 > 9 (careful!)

Answer: x < −3

F. Word Problems

Problem 1 Score 4

A phone plan costs $20 plus $0.05 per text. If the bill was $35, how many texts were sent?

Answer: 300

Problem 2 Score 5

You have $60 and want to buy shirts that cost $12 each. What's the max number of shirts you can buy?

Answer: 5

Problem 3 Score 5

Plan A: $40 to join a club plus $10 each month. Plan B: $20 each month with no joining fee. After how many months do the two plans cost the same?

Answer: 4